

Build Up: Python for Little Geniuses

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Chapter 1

Welcome

Welcome to *Build Up: Python for Little Geniuses!*

In this book, you will learn to write real Python code. Math problems will help you understand each new idea. You will also build your own game, piece by piece, chapter by chapter.

By the end of this chapter, you can...

- Run a line of Python code
- Explain what `print()` does
- Find your way around this book

1.1 Your first code

You do not need to install anything yet. Open this chapter's notebook in Google Colab and run your first line of Python right now.

```
print()
```

```
print("Welcome, Little Genius! You just ran your first line of Python.")
```

Check My Knowledge

1. What word did Python print when you ran the code above?
2. What happens if you change the text inside `print(...)`?

There is no wrong guess here. Try it and see what happens – that is how you learn it, not yet how you fail it.

Good stopping point! Stretch, take a break, and come back whenever you are ready for Chapter 1.

1.2 How to read this book

Every chapter has two tracks that sit side by side:

- **Track A – Foundations.** Short Python lessons. Each one has its own notebook you can run in Google Colab, right in your browser, no install needed.
- **Track B – Build.** One game you build across the whole book. Chapters 0 to 7 work anywhere, Colab included. Chapter 8 is the first time you need to install Python on your own computer, because that is when the game gets a real window you can see and play.

You also have a second no-install option: the **Little Genius Zone** (GitHub Codespaces), a full coding environment in your browser. It also holds optional bonus problems for anyone who finishes a chapter early and wants more of a challenge.

Not everyone needs to read every chapter in the same order. See `learning-paths.tex` for a few different paths through the book, depending on what you already know.

1.3 How we teach

This section is for kids and grown-ups both.

We use a method that starts with something you can touch or picture, and only then moves to code:

1. **Concrete.** A real example, like counting blocks.
2. **Pictorial.** A picture of the same idea, like a diagram.
3. **Abstract.** The actual Python code that does it.

Each chapter also opens with a few guided questions before we explain anything – try to answer them yourself first. Mistakes are part of how this works. We say “not yet,” never “wrong.”

Every chapter ends with two short checks: a self-graded “Check My Knowledge” quiz, and a challenge that mixes in ideas from earlier chapters, so you keep old skills sharp while you learn new ones.

1.4 Taking care of you

A few honest notes before you start:

- **Mistakes are not yet, not wrong.** Every “bug” in your code is a normal part of programming, even for adults who do this every day.
- **There are no streaks to keep here.** This book will never ask you to come back every day or make you feel bad for taking a break.
- **Take real breaks.** Chapters end with a good stopping point on purpose. Stretch, look away from the screen, and come back later.
- **Chapter 8 needs a grown-up.** That is when you install Python on your own computer for the first time. It is a good chapter to do together with a parent or guardian.